Determinants of Inflammatory Markers in a Bi-ethnic Population

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Background Inflammation is a common pathophysiological pathway for a number of chronic diseases, and is strongly influenced by sociodemographic factors and lifestyle. Less is known about factors that may influence the inflammatory response in individuals of distinct ethnic backgrounds. Therefore, this study examined the relationship between ethnicity and blood levels of inflammatory markers in a sample of non-smoking church-goers. Methods In a cross-sectional investigation, 508 men and women (>35 years old, 62% White, 38% Black) participated in the Biopsy-chosocial Religion and Health substudy of the Adventist Health Study 2. The contribution of socioeconomic status (education level and difficulty meeting expenses for basic needs) and health covariates (exercise, vegetarian or other type of diet, body mass index, and presence of inflammatory conditions) toward serum levels of C-reactive protein (CRP), interleukin-6 (IL-6), and tumor necrosis factor-alpha (TNF-α) was assessed with linear regression models. Levels of interleukin-10 (IL-10), an anti-inflammatory marker, were also assessed. Results Blacks showed higher levels of CRP and IL-6 than Whites. Controlling for socio-demographic and health variables attenuated the ethnic difference in CRP while IL-6 levels remained higher in Blacks than in Whites (β=.118; 95% confidence interval=.014–.206; P=.025). Ethnic differences in IL-10 and TNF-α were not found. Vegetarian diet was associated with lower CRP levels while exercise frequency was associated with higher IL-10 levels. Conclusion Higher susceptibility of Blacks to inflammatory diseases may reflect higher IL-6, which could be important in assessing health disparities among Blacks and Whites. Vegetarian diet and exercise may counteract effects of disparities.